IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MICROELECTROMECHANICAL SYSTEMS (MEMS)

Thursday, 16 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer Question 1.

Answer any TWO of the remaining questions (Question 2, 3, and 4)

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : Z. Durrani

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Information for Candidates

The following physical constants may be used:

Permittivity of free space: $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m

The generalised Hooke's Law equations are:

$$E\varepsilon_x = \sigma_x - \nu\sigma_y - \nu\sigma_z$$
$$E\varepsilon_y = \sigma_y - \nu\sigma_z - \nu\sigma_x$$
$$E\varepsilon_z = \sigma_z - \nu\sigma_x - \nu\sigma_y$$

The piezoresistive equations for silicon, referred to axes aligned to the <100> directions, are:

$$E_1/\rho_e = J_1[1 + \pi_{11}\sigma_1 + \pi_{12}(\sigma_2 + \sigma_3)] + J_2\pi_{44}\tau_{12} + J_3\pi_{44}\tau_{13}$$
$$E_2/\rho_e = J_2[1 + \pi_{11}\sigma_2 + \pi_{12}(\sigma_1 + \sigma_3)] + J_1\pi_{44}\tau_{12} + J_3\pi_{44}\tau_{23}$$
$$E_3/\rho_e = J_3[1 + \pi_{11}\sigma_3 + \pi_{12}(\sigma_1 + \sigma_2)] + J_1\pi_{44}\tau_{13} + J_2\pi_{44}\tau_{23}$$

The maximum stress σ_{max} in a square membrane of side L and thickness h subject to a uniformly distributed load p is:

$$\sigma_{max} \approx 0.3\frac{pL^2}{h^2}$$

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This question is compulsory

1. a) Starting from the generalised Hooke's law equations, show that the axial stress-strain relationships in a material where the *z*-component of axial stress is zero can be expressed in the form:

$$\sigma_x = \frac{E}{1-\upsilon^2}\varepsilon_x + \upsilon\frac{E}{1-\upsilon^2}\varepsilon_y$$

$$\sigma_y = \frac{E}{1-\upsilon^2}\varepsilon_y + \upsilon\frac{E}{1-\upsilon^2}\varepsilon_x$$

Give an example of a MEMS structure where these equations would be applicable.

[6]

b) You are required to design a mask for a silicon alignment groove for an optical fibre, as shown in Figure 1.1 below. The fibre has a diameter of 125 μm, and its axis is to be positioned 5 μm above the wafer surface. An optical method is used to stop the etch process when the base width of the trapezoidal trench is reduced to zero. If the KOH etchant you will use is known to have a nominal anisotropy of 400, what width of mask aperture should you use?

[6]

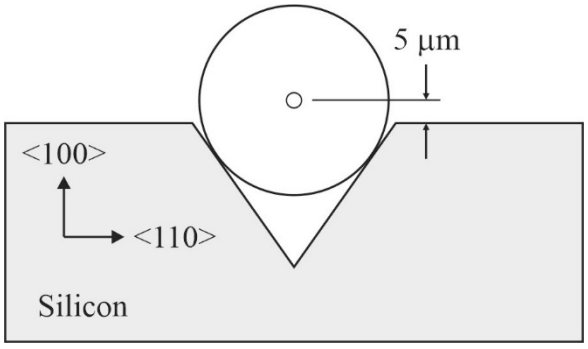


Figure 1.1

c) List the process steps in a typical silicon surface micromachining sequence for a device requiring two mechanical layers. You should identify the materials and processes used at each stage, but you do not need to describe the processes in detail.

[6]

Question 1 continues on the next page.

Question 1 continued.

d) According to elementary diffraction theory, the resolution limit of a projection lithography system with on-axis, plane wave illumination is expected to be:

$$R = k_1 \frac{\lambda}{NA} \quad ; \quad k_1 = 0.5$$

where λ is the optical wavelength and NA is the numerical aperture. Explain briefly, with reference to the concepts of spatial coherence, image modulation depth and resist contrast, why practical lithography systems can achieve better performance than this even without any resolution enhancement techniques being applied.

[5]

e) Starting from the elementary bending equation, derive an expression for the transverse stiffness of a simple cantilever in terms of its dimensions and Young's modulus. Also derive an expression for the maximum tensile strain in terms of the end deflection, and draw a sketch showing where on the deflected cantilever this maximum tensile strain occurs.

[6]

f) The equations of motion for a linear vibratory gyroscope are given by:

$$m\ddot{v}_x + c_x\dot{v}_x + k_x v_x - 2m\Omega\dot{v}_y = F_x$$
$$m\ddot{v}_y + c_y\dot{v}_y + k_y v_y + 2m\Omega\dot{v}_x = F_y$$

Explain briefly the physical significance of the various terms in these equations. Also, assuming that the gyroscope is operated in open loop, and that the resonant frequencies are matched, derive an expression for the ratio of sensed to driven amplitudes when the device is subject to rotation at a constant rate.

[6]

g) Sketch the structure of a typical material bimorph electrothermal actuator and briefly explain its operation. You should include in your answer examples of typical material combinations.

[5]

End of Question 1.

Answer ANY TWO of the remaining questions (Questions 2, 3 and 4)

2. Figure 2.1 shows a silicon micromachined accelerometer comprising a proof mass supported on a hammock suspension. The device is fabricated on a BSOI wafer having a mechanical layer thickness of h = 2 µm and a buried oxide thickness of g = 2 µm. The four suspension beams each have width w = 10 µm and l = length 600 µm, and the proof mass has an area of 1.44 mm^2. The anchors are 100 µm × 100 µm and the oxide is undercut by 10 µm during fabrication.

   a) Suggest a possible fabrication sequence for the device. You should list the process steps involved but you do not need to describe them in detail. Explain why the perforations in the proof mass are needed.

   [6]

   b) Calculate the stiffness of the hammock suspension for out-of-plane displacements of the proof mass. Hence determine the resonant frequency and mechanical sensitivity of the accelerometer under applied accelerations normal to the surface. You should assume the suspension beams are mass-less and ignore the effect of the perforations in the proof mass.

   [10]

   c) The accelerometer is to be read out by measuring the capacitance between the proof mass and the substrate. Estimate the baseline capacitance at zero acceleration and the shifts in capacitance when accelerations of +10 ms^{-2} and –10 ms^{-2} are applied.

   [8]

   d) Suggest a possible readout circuit for the accelerometer, illustrating your answer with a block diagram that shows the key functional blocks.

   [6]

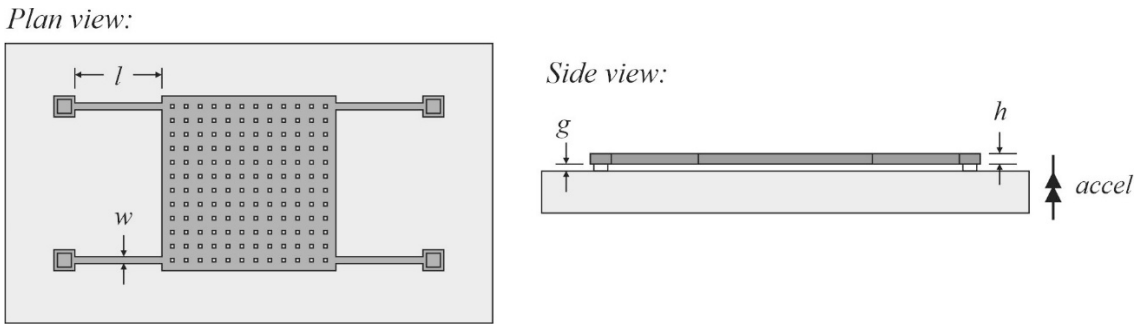


Figure 2.1

You should assume values of 170 GPa and 2330 kg/m^3 for the Young's modulus and density of silicon respectively. The relative permittivity of silicon dioxide is 3.8.

3. a) Describe briefly the piezoelectric and piezoresistive effects and illustrate how they are used in MEMS devices.

[6]

b) A region of silicon is subject to purely axial stresses σ_L and σ_T along the [110] and [1$\bar{1}$0] directions. By considering the equilibrium of a small prismatic element, or otherwise, show that the associated stress components in a coordinate frame aligned to the $< 100 >$ directions are:

$$\sigma_1 = \sigma_2 = \frac{1}{2}(\sigma_L + \sigma_T) \quad ; \quad \tau_{12} = \frac{1}{2}(\sigma_L - \sigma_T)$$

Hence show that the resistance change of a [110]-aligned piezoresistor subject to longitudinal and transverse stresses σ_L and σ_T may be written as:

$$\frac{\Delta R}{R} = \sigma_L \pi_L + \sigma_T \pi_T$$

where $\pi_L = \frac{1}{2}\{\pi_{11} + \pi_{12} + \pi_{44}\}$ and $\pi_T = \frac{1}{2}\{\pi_{11} + \pi_{12} - \pi_{44}\}$, and π_{ij} are the standard piezoresistive coefficients.

[12]

c) A bulk micromachined pressure sensor has a square silicon membrane of side length 2 mm and thickness 10 μm. The membrane stress is monitored using a small piezo-resistive bridge positioned at the mid-point of one of the membrane edges. The resistors in the bridge are matched and are aligned either parallel or perpendicular to the membrane edge. What will be the sensitivity of the device (in Volts per Pascal) if it is operated with the bridge connected to a 5 V supply?

[12]

Assume the following values for the piezoresistive coefficients of the silicon (all in units of 10^{-11} Pa^{-1}): $\pi_{11} = 6.6$, $\pi_{12} = -1.1$, $\pi_{44} = 138.1$. Poisson's ratio for the relevant directions is $\nu = 0.064$.

4. Figure 4.1 shows the model for an out-of-plane electrostatic actuator comprising a moveable plate on an elastic suspension and a fixed lower plate. The electrode gap at zero applied voltage is g_0, and spacers either side of the lower plate set the minimum gap, s.

   a) i) Derive an expression for the total upward force F on the moveable plate, and show that it can be expressed in normalised form as:

$$f = 1 - u - \frac{a}{u^2}$$

   where $f = F/(kg_0)$ is the normalised force, $u = g/g_0$ is the normalised gap, and $a = A\varepsilon_0 V^2/(2kg_0^3)$. [6]

   ii) Sketch the variation of f with u for several values of a, and hence explain the origin of snap-down instability. Also sketch the variation of gap with applied voltage. [8]

   b) State the conditions that apply at the point of snap-down, and hence derive an expression for the critical voltage, V_P, at which snap-down occurs. [4]

   c) It is suggested that snap down can be avoided by placing a fixed capacitor C in series with the actuator rather than driving it directly from a voltage source. Show that in this case the normalised force becomes:

$$f = 1 - u - \frac{a}{(u+b)^2}$$

   where $b = C_0/C$, with C_0 being the rest capacitance of the actuator at zero applied voltage. By considering the snap-down condition in this case, show that the actuator will not exhibit snap-down provided the series capacitance satisfies:

$$C \leq \frac{C_0}{(2 - 3s/g_0)}$$

   [8]

   d) If C has the maximum allowed value derived in Part c), and $s << g_0$, show that the voltage required to bring the moveable plate to the bottom of its travel is approximately $V = 3\sqrt{3}V_P$. [4]

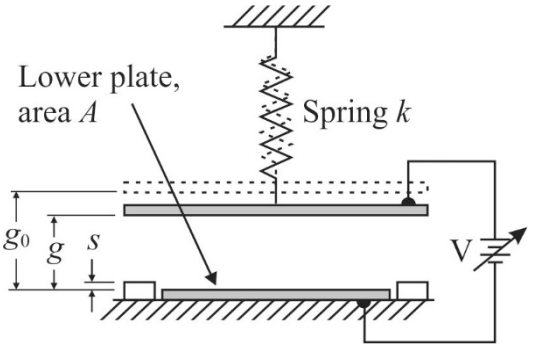


Figure 4.1